

1. Inline E=mc²: $E = mc^2$

2. Multiple inline: $x + y = z$

3. Display fraction:

$$\frac{a}{b}$$

4. Quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

5. Superscript: x^2

6. Subscript: a_1

7. Sub+Super: x_i^2

8. Square root: $\sqrt{x}$

9. Summation:

$$\sum_{i=1}^n i^2$$

10. Integral:

$$\int_0^1 x^2 dx$$

11. Matrix:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

12. Accent hat: $\hat{x}$

13. Function sin: $\sin(x)$

14. Overbar: $\bar{x}$

15. Lower limit: $\lim_{n \rightarrow \infty}$

16. Group char: $\underbrace{a + b + c}$

17. Delimiter: $(x + y)$

18. Mixed paragraph with math and text.